

EE5750 Quantum Error Correction

Homework 7

Due: 17 Sep 2026

As part of the Homework assessment, provide solutions to only the questions marked as **(*Homework*)**. The rest of the questions are practice problems.

1. **(*Homework*)** *Density operators span the entire matrix space*

The class of density operators $\mathcal{D}(\mathbb{C}^d)$ associated with a vector space $\mathbb{C}^d$ are the operators that are positive and have trace 1. We will show in the present question that any operator in $\mathbb{C}^{d \times d}$ can be expressed as a linear combination of density operators. This observation will be used in the next question.

This question uses some of the problems from earlier Homeworks. This is the sequence of arguments that you need to prove

- (a) Every positive operator is a scalar times a density operator.
- (b) Every Hermitian operator is the difference between two positive operators.
Hint: Use the spectral decomposition.
- (c) Every operator (matrix in $\mathbb{C}^{d \times d}$) is a linear combination of two Hermitian operators.
- (d) Conclude that every operator is a linear combination of density operators.

2. **(*Homework*)** *Extending the domain of a quantum channel from $\mathcal{D}(\mathbb{C}^d)$ to $\mathbb{C}^{d \times d}$*

This question is based on Appendix B of Mark M Wilde's *Quantum Information Theory*.

Context. From an operational viewpoint a quantum channel is a mathematical abstraction of a physical process that transforms an input quantum state to an output quantum state. In this physical sense, a quantum channel is a map from $\mathcal{D}(\mathbb{C}^{d_1})$ to $\mathcal{D}(\mathbb{C}^{d_2})$, where d_1 and d_2 are the dimensions of the state spaces of the input and output registers, respectively, that is, formally a quantum channel $\mathcal{E}$ is a map

$$\mathcal{E} : \mathcal{D}(\mathbb{C}^{d_1}) \rightarrow \mathcal{D}(\mathbb{C}^{d_2}). \quad (1)$$

It is reasonable to impose *convex linearity* on $\mathcal{E}$, that is

$$\mathcal{E}(p_1 \rho_1 + p_2 \rho_2) = p_1 \mathcal{E}(\rho_1) + p_2 \mathcal{E}(\rho_2) \quad (2)$$

where $p_1, p_2 \geq 0$ and $p_1 + p_2 = 1$ and $\rho_1, \rho_2 \in \mathcal{D}(\mathbb{C}^{d_1})$. This follows from the idea that if the input is chosen between ρ_1, ρ_2 with probabilities p_1, p_2 then the outputs $\mathcal{E}(\rho_1)$ and $\mathcal{E}(\rho_2)$ occur with probabilities p_1, p_2 .

Question. We now argue that the domain of any function $\mathcal{E}$ as in (1) that satisfies (2) can be extended to a linear map

$$\tilde{\mathcal{E}} : \mathbb{C}^{d_1 \times d_1} \rightarrow \mathbb{C}^{d_2 \times d_2}.$$

In other words, given a convex-linear map $\mathcal{E}$ whose domain is density operators there exists a linear map $\tilde{\mathcal{E}}$ whose domain is *all* operators such that

$$\tilde{\mathcal{E}}(\rho) = \mathcal{E}(\rho) \text{ for all density operators } \rho \in \mathcal{D}(\mathbb{C}^{d_1}), \quad (3)$$

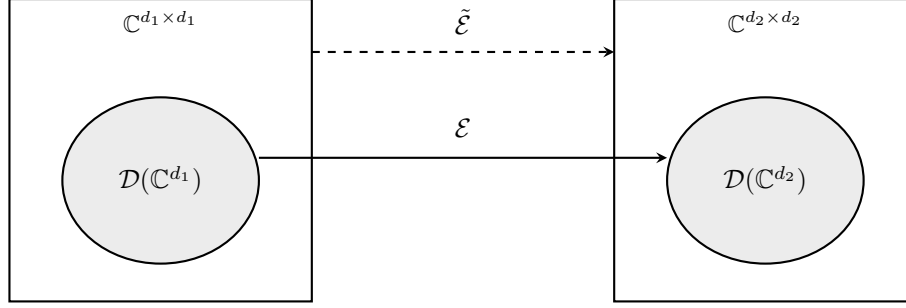


Figure 1: The quantum channel $\mathcal{E}$ acts on the convex set of density operators $\mathcal{D}(\mathbb{C}^{d_1}) \subset \mathbb{C}^{d_1 \times d_1}$, mapping it into $\mathcal{D}(\mathbb{C}^{d_2}) \subset \mathbb{C}^{d_2 \times d_2}$. Its linear extension $\tilde{\mathcal{E}}$ acts on the full operator space $\mathbb{C}^{d_1 \times d_1}$, mapping it into $\mathbb{C}^{d_2 \times d_2}$, and agrees with $\mathcal{E}$ on density operators.

see Figure 2. This is the mathematical sense in which it is okay to talk about a quantum channel acting on an arbitrary operator (even if that operator is not a density matrix).

We will now assume that a map $\mathcal{E}$ satisfying (1) and (2) is given to us. We will constructively build its linear extension $\tilde{\mathcal{E}}$, satisfying (3), using the same logical sequence as in question 1.

- Since $\tilde{\mathcal{E}}$ has to be linear, what must be the value of $\tilde{\mathcal{E}}(0)$.
Remark: The all-zero matrix O does not lie in $\mathcal{D}(\mathbb{C}^{d_1})$ (why?).
- Since every positive operator is a scalar times a density operator, how will you define $\tilde{\mathcal{E}}(N)$ for a positive operator N ?
- Now show how $\tilde{\mathcal{E}}$ will act on Hermitian operators.
- Finally, consider an arbitrary operator as input to $\tilde{\mathcal{E}}$.

Remark: Although we did not prove it here, this construction of $\tilde{\mathcal{E}}$ from $\mathcal{E}$ is *consistent* (that is, every input matrix is assigned a unique output matrix, the construction does not contradict itself) and is *unique* (that is, every $\mathcal{E}$ has a unique linear extension $\tilde{\mathcal{E}}$).

3. (*Homework*) Choi representation of quantum channels

Context. In many instances, a quantum channel is specified to us only in terms of how it acts on density operators. We might need to use the idea in question (2) to mathematically characterize its action on all inputs; for instance to find the Choi representation of a quantum channel.

In this question we will not distinguish a physical quantum channel and its linearization, and use the notation $\mathcal{E}$ for both. This notation is consistent with our lecture notes where we use $\mathcal{E} : \mathbb{C}^{d_1 \times d_1} \rightarrow \mathbb{C}^{d_2 \times d_2}$ to denote a quantum channel or a CPTP map.

Question. Consider the case $d_1 = d_2 = 2$. Consider the quantum channel $\mathcal{E}$ that acts on density operators as below

$$\mathcal{E}(\rho) = \frac{1}{2}\rho + \frac{1}{2}|0\rangle\langle 0| \text{ for all } \rho \in \mathcal{D}(\mathbb{C}^2).$$

$\mathcal{E}$ can be realized by choosing between the input state ρ and the pure state $|0\rangle\langle 0|$ with equal probability.

- Consider the four operators $|i\rangle\langle j|$, $i, j \in \{0, 1\}$. Which of them belong to $\mathcal{D}(\mathbb{C}^2)$?
- Determine $\mathcal{E}(|i\rangle\langle j|)$ for all i, j .

Remark: You will have to rely on question (2) for this.

- What is J and what is the Choi rank of $\mathcal{E}$, i.e., what is $\text{rank}(J)$?
- Find a Kraus representation of $\mathcal{E}$ in which the number of Kraus operators equals the Choi rank.
Hint: You must factorize J as $BB^\dagger$ where the number of columns of B equals $\text{rank}(J)$. You can use a computer for this if you have to.

4. **(*Homework*)** A quantum channel with maximum Choi rank

Consider a quantum channel $\mathcal{E}$ whose Choi representation J is a scalar times the identity matrix, i.e.,

$$J = \alpha I_{d_1 d_2 \times d_1 d_2}.$$

The rank of J is $d_1 d_2$ (assuming $\alpha \neq 0$), which is the maximum possible rank among Choi matrices since they are matrices of size $d_1 d_2 \times d_1 d_2$.

- (a) Find the value of α .
- (b) Find $\mathcal{E}(|i\rangle\langle j|)$ for all $i, j \in \{0, 1, \dots, d_1 - 1\}$.
- (c) Find a Kraus representation for $\mathcal{E}$.
- (d) Find a simplified expression for $\mathcal{E}(\rho)$ for all $\rho \in \mathcal{D}(\mathbb{C}^{d_1})$.

5. Argue that $\frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |i\rangle \otimes |i\rangle$ is an entangled state.

6. *Maximally entangled quantum states*

Consider a composite register AB where the state spaces of both A and B are $\mathbb{C}^d$. A pure state ψ on AB is called *maximally entangled* when the states of the individual registers A and B are both $\frac{1}{d} I_{d \times d}$. Recall that I/d is called the *maximally mixed state*.

Use Schmidt decomposition to identify *all* maximally entangled states.

Remark: Maximally entangled states are an important resource in quantum communication and computing. For instance, they are used in quantum teleportation and in superdense coding.

Remark: These states are the solution to Problem 5 in Exam 2.

7. **(*Homework*)** Verifying unitary freedom in Kraus form

Suppose $\mathcal{E}(N) = \sum_{k=1}^t V_k N V_k^\dagger$ and $E_j = \sum_k u_{j,k} V_k$ for some unitary matrix U . Directly verify

$$\sum_{k=1}^t V_k N V_k^\dagger = \sum_{j=1}^t E_j N E_j^\dagger$$

by substituting the expression for E_j on the RHS.